

UNIT - 3

- Prime Numbers :
 - Importance
 - Coprime
 - Euler's Phi Function
 - Role in RSA
- Chinese Remainder Theorem :
 - Definition
 - Formula
 - Steps to Solve
- Asymmetric Key Cryptography :
 - Idea
 - Symmetric vs Asymmetric
 - Working (Flow)
 - Encryption / Decryption
 - One-Way Function (OWF)
 - Trapdoor OWF
 - Knapsack Cryptosystem
- Rabin Cryptosystem :
 - Idea
 - Encryption / Decryption
 - Key Generation
 - Difference from RSA
- Key Management :
 - What ?
 - N^2 Problem
 - KDC (Key Distribution Center)
 - Session Key
- Symmetric Key Agreement :
 - Idea
- Diffie-Hellman Key Exchange :
 - Idea
 - Steps
 - Attacks
 - Security Conditions

UNIT - 4

- Hash Functions :
 - Definitions
 - Properties
 - Applications
 - Iterated Hash Functions
 - Merkle-Damgard Scheme
- MD5 :
 - Features
 - Steps
 - Limitations
- SHA (Secure Hash Algorithm) :
 - Versions
 - SHA-512 : Key Points
- Digital Signature :
 - Definition
 - Process
 - Digital vs Conventional
 - Services Provided
- Attacks on Digital Signature :
 - Types
 - Forgery Types
- RSA Digital Signature :
 - Idea
 - Equations
- DSS (Digital Signature Standard) :
 - Key Points
 - Signing
 - Verification